

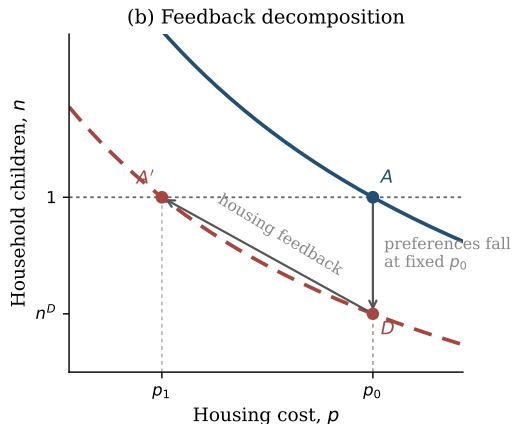
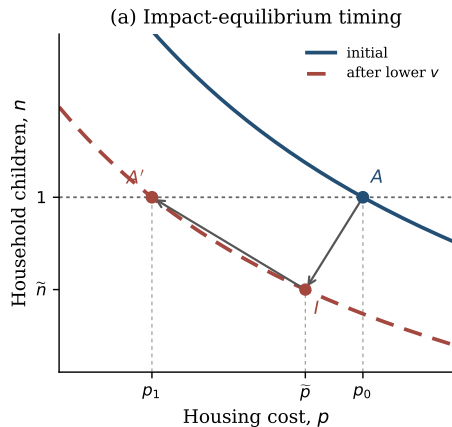
Alternative Views of the Toy-Model Transition

Figures for discussion

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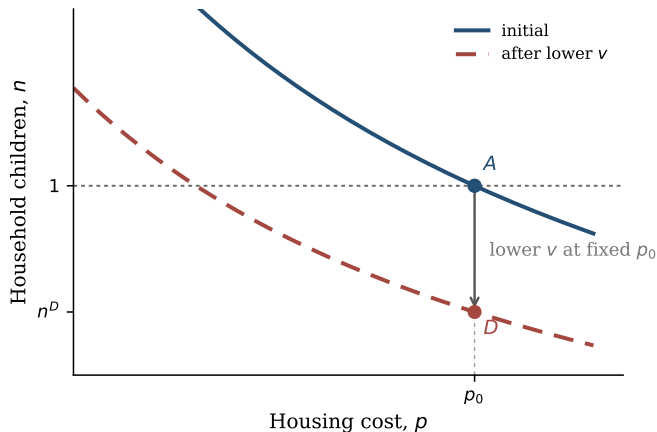
Two ways to organize the same feedback



Impact timing: I is the market-clearing impact point with inherited cohorts.

Feedback decomposition: D holds $p = p_0$; the move to A' shows how falling housing costs prevent the no-feedback demographic contraction from continuing.

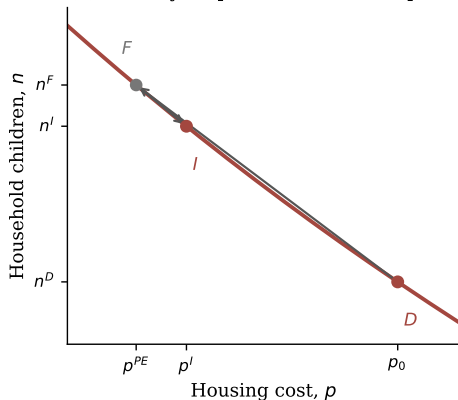
Alternative 1: the primitive fertility shock



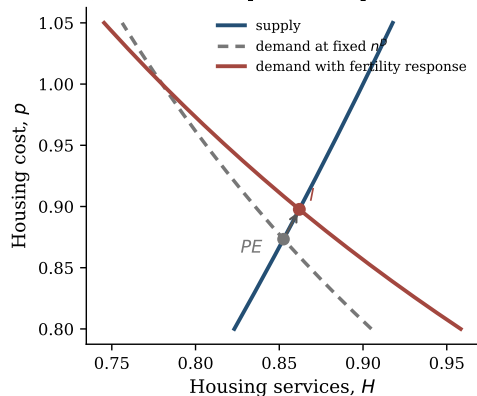
- **Primitive:** $v_0 \rightarrow v_1 < v_0$.
- **Hold** $p = p_0$: fertility falls from 1 to n^D .
- **Point D :** prices and cohort sizes have not yet adjusted.

Alternative 2: fast equilibrium feedback

(a) Fertility responds to the lower price



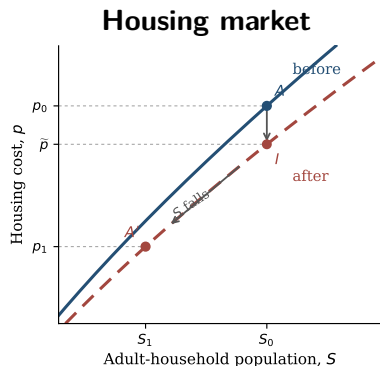
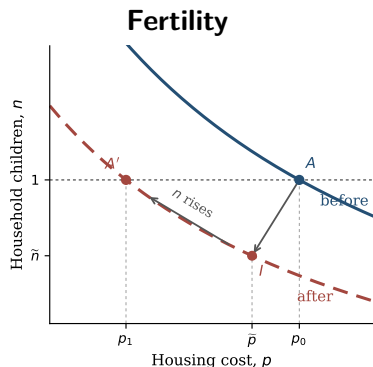
(b) Child demand pushes the price back up



Holding $n = n^D$ gives the mechanical price p^{PE} . At that price households want $n^F > n^D$, so child-related demand pushes the price back to p^I and fertility down to n^I :

$$p^{PE} < p^I < p_0, \quad n^D < n^I < n^F < 1.$$

Alternative 3: slow cohort adjustment



- At $I, n^I < 1$: smaller cohorts gradually reduce housing demand and p .
- Lower p raises fertility along the new curve until the economy reaches A' with $n = 1$ and $S_1 < S_0$.

The toy model now has a genuine transition

After the permanent fall in the value of children, $v_0 \rightarrow v_1 < v_0$, each date is determined by

$$Y_{t+1} = n(p_t; v_1)Y_t, \quad O_{t+1} = Y_t, \\ Y_t h_Y(p_t; v_1) + O_t h_O(p_t) = H^s(p_t).$$

The first line advances the two cohorts; the second clears the housing market given their current sizes.

With total adult-household population $S_t = Y_t + O_t$,

$$S_{t+1} - S_t = Y_t \left(n_t - \frac{O_t}{Y_t} \right).$$

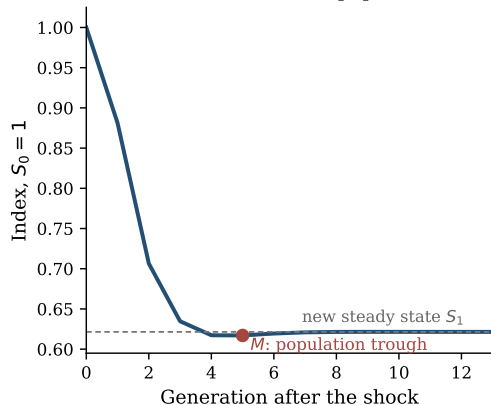
Population therefore bottoms when fertility crosses the inherited old-to-young cohort ratio, not necessarily when fertility crosses one.

$$A \longrightarrow I \longrightarrow P \longrightarrow M \longrightarrow A'.$$

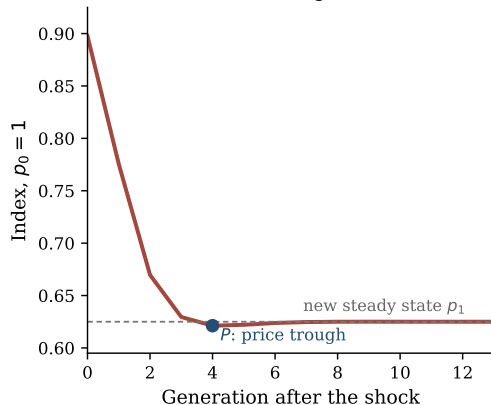
A : initial steady state; I : impact equilibrium; P : price trough; M : population trough; A' : new steady state.

Population and housing costs undershoot the new steady state

(a) Adult-household population

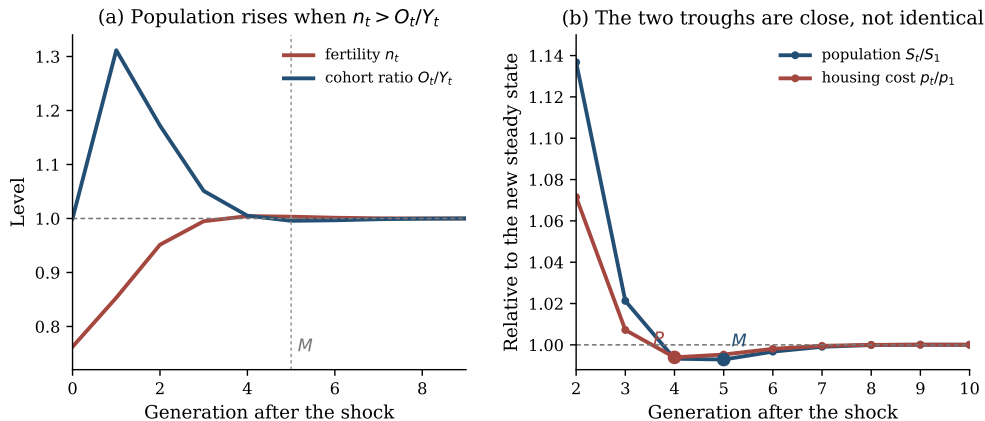


(b) Housing cost



In this illustration, housing cost reaches its minimum at P in generation 4 and population reaches its minimum at M in generation 5. Both then recover slightly toward A' .

The price trough comes just before the population trough



Population keeps falling while $n_t < O_t/Y_t$, even after fertility has returned above one. The two troughs need not coincide because housing demand also depends on the composition of the population between young and old households.